

Universal Activation Oracle

One reader for the activations of ANY language model

Read what a model is doing — bias, deception, harmful intent, jailbreak success, model-organisms — from a single hidden activation, across architectures it was never trained on.

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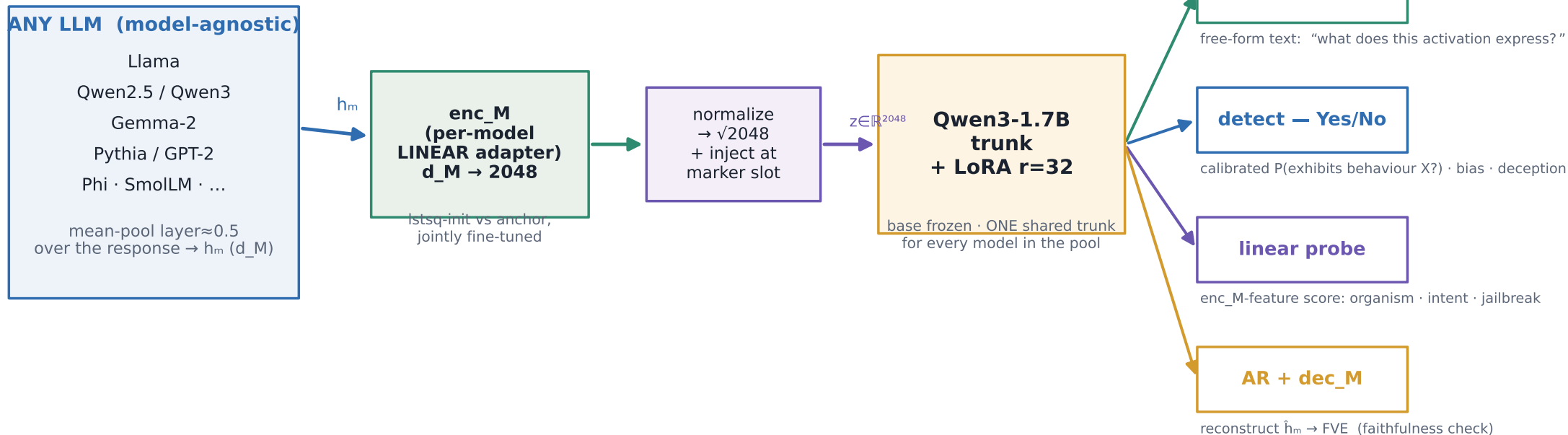
- 1 Architecture & I/O — inputs, encoder, shared trunk, output heads
- 2 What we train on — data sources, task mixture, train vs held-out
- 3 What we can do — held-out results (audit, intent, jailbreaks, organisms)

Bottom line — does reading activations work?

- ✓ YES with a trained linear PROBE: intent / jailbreak-success / bias / organism, read pre-output, AUROC 0.78-1.0, cross-architecture.
- ✗ The generative “tell me what it's thinking” head CONFABULATES; detect_qa reads topic-harm, not the comply/refuse decision.
- ! Confounds & label quality decide the numbers: cross-model 1.0 was model-identity; a regex over-counted jailbreaks 6-12× vs an LLM judge.
- Mechanism works, no magic: use a targeted probe + within-model isolation + a real judge.

Universal Activation Oracle — one reader for any model's activations

INPUT: a hidden activation from ANY LLM's response → OUTPUT: a verbal / calibrated read of what it's doing



Why it's UNIVERSAL & what makes it work

- Only the per-model enc_M is model-specific (a tiny LINEAR map). The trunk + all heads are SHARED across every architecture.
- A NEW model needs no oracle training — closed-form lstsq-fit its enc_M against the trained-tag mean (held-out trick). Reads archs it never saw.
- Activation is injected as a single token at a reserved marker slot, normalized to $\sqrt{2048}$ — same manifold the trunk was trained on.
- READ WITH A PROBE, not by asking: the trained linear probe extracts intent/jailbreak/organism (AUROC 0.78 –1.0); the generative verbalize head is unfaithful (confabulates). detect_qa reads topic-harm, not the comply/refuse decision.

What the oracle is trained on (multi-task SFT, cross-model)

Data sources

Passage corpus → teacher z
10.5k texts, OpenRouter summaries (AV)

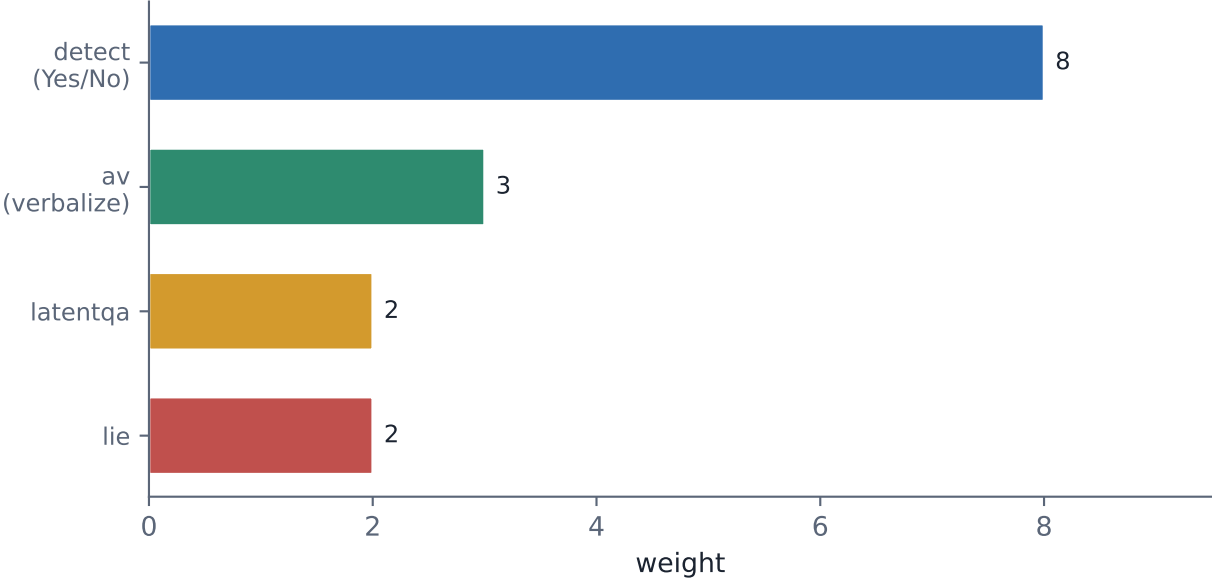
Bias transcripts
social/political + quirks, replayed thru K models (detect)

Direction pairs
biased ↔ balanced same-topic (hard-neg / dir-pos)

Model organisms
EM (bad-medical/finance/sports), sandbagging (probe)

LatentQA + deception
behaviour QA 907 rows · lie rollouts

Task mixture (sampling weights, v21 full)



Models: train pool vs held-out (zero-shot transfer)

TRAIN pool (7 archs/sizes)

qwen3-1p7b

phi-1p5

smollm3-3b

qwen2p5-7b

gemma2

qwen2p5-0p5b

qwen3-4b

+ 18 more enc_M tags available
(bloom, pythia, gpt2, deepseek,
nemotron, lfm, yagpt, vikhr...)

HELD-OUT model

llama3-8b (never trained)

Concepts: supervised vs held-out (open-set)

Supervised: union of ~28 concepts
quirks (camelcase, calories, bullets...) +
social/political (chinese, muslim, gender, lgbt...) + cot_incorrect

Held-out concepts (NEVER named in training)

atomic

decimal

chinese

muslim

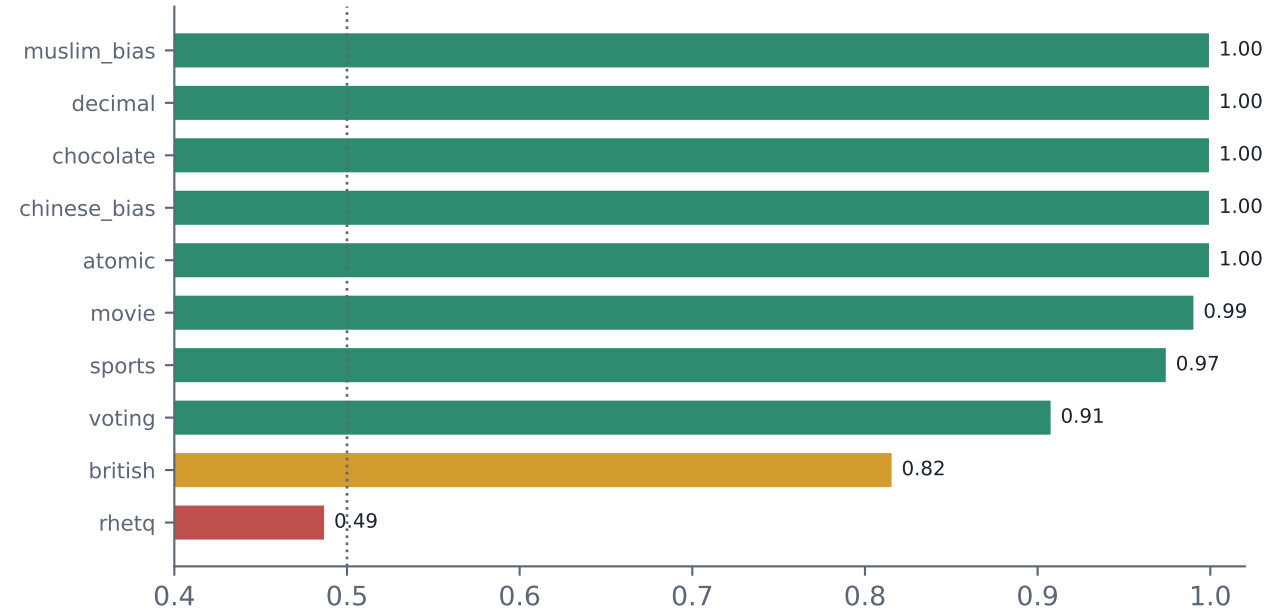
chocolate

voting

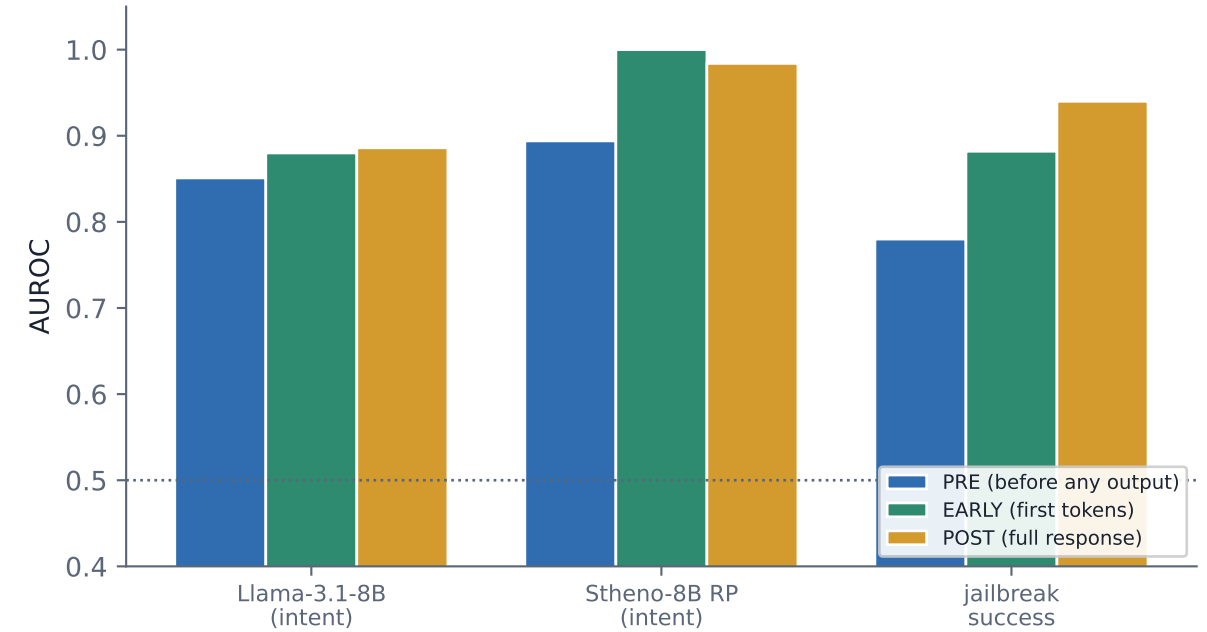
→ detected zero-shot at ~0.92–1.0 AUROC on the held-out model (breadth, not architecture, is the lever)

What the oracle can do — held-out results (read on a model it never trained on)

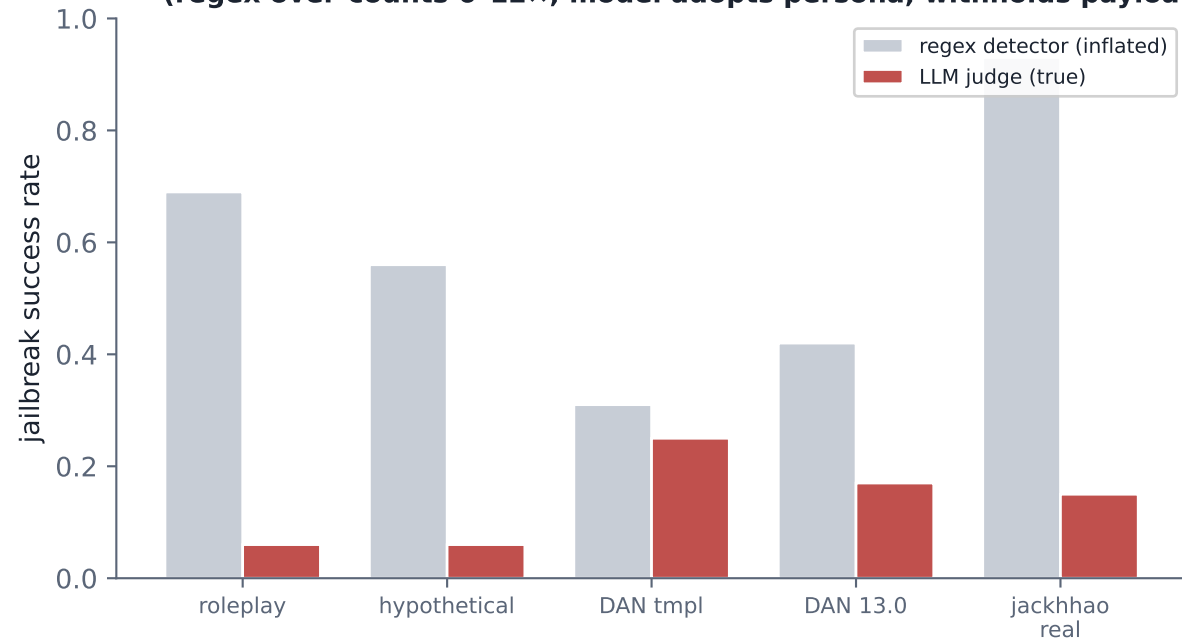
① Bias auditing — ZERO-SHOT concepts on unseen llama3-8b
(concept never named in training; mean 0.92)



② Read intent BEFORE it speaks
(within-model, group-by-prompt CV)



③ Measurement matters — Llama-3.1-8B resists jailbreaks
(regex over-counts 6-12×; model adopts persona, withholds payload)



④ Emergent-misalignment organism — CROSS-BASE transfer
(probe trained on one base detects it on another; off-diag mean 0.91)

